

Exercise sheet 2.

Name _____

Exercise **1** **2** **3** **4** **Σ**

Points

Exercise group (tutor's name) _____

Deadline: **Friday, 5.11.2021, 16:00.**

Please use this page as a cover sheet and enter your name and tutor in the appropriate fields. Please staple your solutions to this cover sheet.

The following exercises are about *functors*. The relevant reading material is Section 1.3 from Emily Riehl's book *Category Theory in Context*.

Exercise 1. Find an example of a functor $F: C \rightarrow D$ for which the objects and morphisms in the image do not define a subcategory of D .

Exercise 2. Let $F: D \rightarrow C$ and $G: E \rightarrow C$ be functors. Show that there is a category – called the *comma category* $F \downarrow G$ – which has

- as objects, triples $(d \in D, e \in E, f: Fd \rightarrow Ge \in C)$;
- as morphisms $(d, e, f) \rightarrow (d', e', f')$, a pair of morphisms $h: d \rightarrow d'$ and $k: e \rightarrow e'$ such that the following square commutes:

$$\begin{array}{ccc} Fd & \xrightarrow{f} & Ge \\ Fh \downarrow & & \downarrow Gk \\ Fd' & \xrightarrow{f'} & Ge' \end{array}$$

Exercise 3. What is a functor between two groups, viewed as categories?

What is a functor between two preorders, viewed as categories?

Exercise 4. Functoriality sometimes depends on the categories that are chosen. Consider three standard constructions from group theory. Let G be a group. Its *centre* is

$$Z(G) = \{h \in G : hg = gh \text{ for all } g \in G\}.$$

Its *commutator subgroup* $[G, G] \subseteq G$ is the subgroup generated by all “commutators” $ghg^{-1}h^{-1}$, $g, h \in G$. (This subgroup is normal, and is the smallest normal subgroup N so that G/N is Abelian.) Its automorphism group $\text{Aut}(G)$ is the group of all group isomorphisms from G to itself. Let

$$\mathbf{Groups} \supseteq \mathbf{Groups}_{\text{epi}} \supseteq \mathbf{Groups}^*$$

be the categories of groups with *all* group homomorphisms, *surjective* group homomorphisms, and *bijective* group homomorphisms as arrows. For each of the three constructions $Z(G)$, $[G, G]$, $\text{Aut}(G)$, does it come from a functor $\mathbf{Groups} \rightarrow \mathbf{Groups}$, $\mathbf{Groups}_{\text{epi}} \rightarrow \mathbf{Groups}$, or $\mathbf{Groups}^* \rightarrow \mathbf{Groups}$?

(A group homomorphism is an epimorphism in $\mathbf{Groups}$ if and only if it is surjective. The proof of this theorem is due to Otto Schreier and uses amalgamated free products of groups. You need not prove this!)